

PBS / MegaRAID

Failed HDD Replacement

Junior Runbook - Safe Hot-Swap Procedure

Platform	Proxmox Backup Server (PBS)
RAID controller	Broadcom / AVAGO MegaRAID SAS 9361-4i
CLI tool	/opt/MegaRAID/storcli/storcli64
Example failed disk	Controller c0 / Enclosure e4 / Slot s16

WARNING: Never remove a disk only because an LED is red. Identify the failed disk in StorCLI and confirm the physical bay with 'locate'.

1. Find the failed disk

```
/opt/MegaRAID/storcli/storcli64 /c0/eall/sall show
```

Look for a drive in UBad (Unconfigured Bad) state. Example:

```
EID:SlT    State    DG
4:16      UBad      -
```

DG = - means the drive is not currently assigned to a Drive Group.

2. Check RAID status

```
/opt/MegaRAID/storcli/storcli64 /c0/vall show
```

Verify the Virtual Drives before touching hardware. A healthy example is:

```
0/0  RAID5  Optl
1/1  RAID5  Optl
2/2  RAID5  Optl
3/3  RAID5  Optl
```

WARNING: If another Virtual Drive is Dgrd, Pdgd or OfLn, stop. Do not remove another disk until the RAID condition is understood.

3. Identify the physical bay

Use the enclosure and slot found in Step 1. Example for e4:s16:

```
/opt/MegaRAID/storcli/storcli64 /c0/e4/s16 start locate
```

Identify the bay, then stop Locate:

```
/opt/MegaRAID/storcli/storcli64 /c0/e4/s16 stop locate
```

Example 24-bay map used during this maintenance:

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16 FAILED
17	18	19	20
21	22	23	24

4. Replace the failed HDD

When the failed disk is confirmed as UBad, DG = -, and the correct bay has been identified, remove only that hot-swap drive.

Wait 15-30 seconds, then perform one safety check:

```
/opt/MegaRAID/storcli/storcli64 /c0/vall show
```

The Virtual Drives should remain in the expected state. Insert the compatible replacement disk into the same bay.

WARNING: Do not use 'set good', 'force online', 'set offline', 'clear foreign' or manual rebuild commands unless the actual drive state specifically requires them and the procedure has been reviewed.

5. Check the replacement disk

Wait 20-30 seconds after insertion, then check the replacement:

```
/opt/MegaRAID/storcli/storcli64 /c0/e4/s16 show all
```

Check model, capacity, sector size, link speed, SMART status and error counters. In this case the controller automatically changed the replacement from UGood to Cpybck.

6. Monitor automatic CopyBack

If the new disk is shown as Cpybck, do not start a rebuild manually. Monitor the automatic CopyBack:

```
/opt/MegaRAID/storcli/storcli64 /c0/e4/s16 show copyback
```

Example:

```
/c0/e4/s16 0% In progress Estimated Time Left: 12 Days 19 Hours
```

WARNING: During CopyBack, do not remove the source or destination drive. Avoid unnecessary reboot/shutdown until the operation completes.

7. Final check after CopyBack

```
/opt/MegaRAID/storcli/storcli64 /c0/vall show
```

Confirm all Virtual Drives are Optl. Then check the controller and drive states:

```
/opt/MegaRAID/storcli/storcli64 /c0 show  
/opt/MegaRAID/storcli/storcli64 /c0/eall/sall show
```

Verify which drive is now Online and whether the original hot spare has returned to a spare state. Do not assume the final spare role; confirm it in StorCLI.

Optional - Silence the MegaRAID alarm on PBS

If the controller buzzer is sounding during a known recovery operation, first check the alarm:

```
/opt/MegaRAID/storcli/storcli64 /c0 show alarm
```

Silence the current alarm:

```
/opt/MegaRAID/storcli/storcli64 /c0 set alarm=silence
```

Prefer silence over disabling the alarm. Avoid:

```
/opt/MegaRAID/storcli/storcli64 /c0 set alarm=off
```

because 'off' disables the alarm function rather than only silencing the current audible alarm.

Quick reference

Task	Command
List drives	/opt/MegaRAID/storcli/storcli64 /c0/eall/sall show
Check virtual drives	/opt/MegaRAID/storcli/storcli64 /c0/vall show
Locate drive	/opt/MegaRAID/storcli/storcli64 /c0/eX/sY start locate
Stop locate	/opt/MegaRAID/storcli/storcli64 /c0/eX/sY stop locate
Drive details	/opt/MegaRAID/storcli/storcli64 /c0/eX/sY show all
CopyBack progress	/opt/MegaRAID/storcli/storcli64 /c0/eX/sY show copyback
Alarm status	/opt/MegaRAID/storcli/storcli64 /c0 show alarm
Silence alarm	/opt/MegaRAID/storcli/storcli64 /c0 set alarm=silence

Rule: identify -> verify RAID -> locate -> replace -> verify -> monitor.